

Homework 2

Advanced Applied Math (MAE5003)

Posted: May. 1, 2026

Due Date: May. 12, 2026

Problem 1

Matched asymptotic expansion for the boundary layer problem

For $\epsilon \ll 1$,

$$\begin{aligned}\epsilon y'' + 2y' + 2y &= 0, \\ y(0) &= 1, y(1) = 1.\end{aligned}\tag{1}$$

(a) Since Eq. 1 is a constant coefficient ODE, use the exponential ansatz (Lecture 9) to solve for its exact solution, regardless of the size of ϵ .

(b) For small ϵ , construct a 1-term approximation of the solution of Eq. 1:

- Where is the boundary layer and why?
- For the region far from the boundary layer, construct the outer solution.
- For the region of boundary layer, construct the inner solution (lowest order). Identify the size of the boundary layer.
- Match the inner and outer solutions in the transition region to obtain a consistent solution. Plot the approximate solution and compare with the exact solution.

Problem 2

Boundary layer versus WKB

For Eq. 1, if the first order derivative term is missing, then it becomes

$$\begin{aligned}\epsilon y'' + Cy &= 0, \\ y(0) &= 1, y(1) = 1,\end{aligned}\tag{2}$$

which is naturally solved using WKB. Discuss what happens if one “forces” a boundary-layer asymptotic matching method for Eq. 2. Separate the cases of the constant $C > 0$ and $C < 0$.

Problem 3

Consider the nonlinear ODE for $0 < x < 1$

$$\begin{aligned}\epsilon y'' + y(1 - y^2)y' - y &= 0, \\ y(0) &= \alpha, y(1) = \beta.\end{aligned}\tag{3}$$

(a) Convert the second-order Eq. 3 into two first-order equations. That is to say, turn Eq. 3 into a dynamical system. Use `StreamPlot` in `Mathematica` to sketch the phase space for this dynamical system (use a small ϵ , e.g., 0.1).

(b) For $\alpha = 3/2$, $\beta = -3/2$, discuss the “fast” and “slow” regions of the phase space. Qualitatively sketch the solution in the x - y space.

(c) Repeat (b) for $\alpha = 1/2$, $\beta = 2$. Identify the location of the boundary layer(s).